

Why deep-sea divers use air diluted with helium gas in their air tanks?

The professional deep-sea divers carry a compressed air tank for breathing at high pressure under water. The normal compressed air contains nitrogen and oxygen and these gases are not very soluble in blood and other body fluids at normal pressure. As the pressure at that depth is far greater than the surface atmospheric pressure, more nitrogen dissolves in the blood and other body fluids when the diver breathes from tank. When the diver ascends to the surface, the pressure decreases, the dissolved nitrogen comes out of the blood and other body fluids quickly forming bubbles in the blood stream. These bubbles restrict blood flow, affect the transmission of nerve impulses and can even burst the capillaries or block them. This condition is called "the bends", which are painful and dangerous to life. Now a days, to avoid such dangerous condition, the professional divers, use air diluted with helium gas (about 11.7% Helium, 56.2% Nitrogen and 32.1% Oxygen), because of lower solubility of helium in the blood than nitrogen. Moreover, because of small size of helium atoms they can pass through cell walls without damaging them. The excess oxygen dissolved in the blood is used in metabolism and does not cause the condition of bends

9.5.1 Limitations of Henry's law

- Henry's law is applicable at moderate temperature and pressure only.
- Only the less soluble gases obeys Henry's law
- The gases reacting with the solvent do not obey Henry's law. For example, ammonia or HCl reacts with water and hence does not obey this law.

$$NH_3 + H_2O \leftrightarrows NH_4^+ + OH^-$$

- The gases obeying Henry's law should not associate or dissociate while dissolving in the solvent.

Example Problem 2:

0.24 g of a gas dissolves in 1 L of water at 1.5 atm pressure. Calculate the amount of dissolved gas when the pressure is raised to 6.0 atm at constant temperature.

$p_{solute} = K_H x_{solute\ in\ solution}$

At pressure 1.5 atm,

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